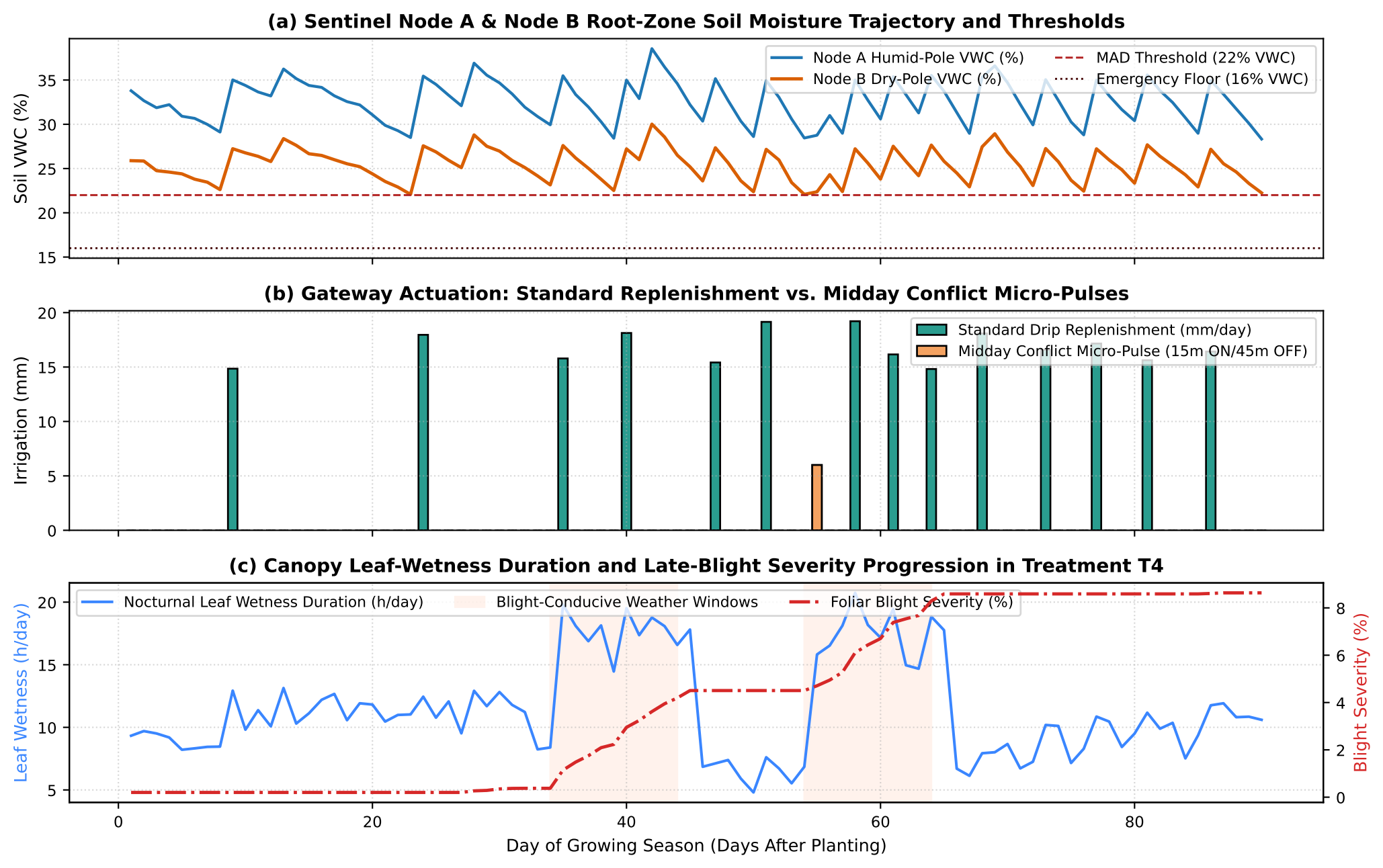
(a) Sentinel Node A & Node B Root-Zone Soil Moisture Trajectory and Thresholds
Soil VWC (%)
Node A Humid-Pole VWC (%)
Node B Dry-Pole VWC (%)
MAD Threshold (22% VWC)
Emergency Floor (16% VWC)
(b) Gateway Actuation: Standard Replenishment vs. Midday Conflict Micro-Pulses
Irrigation (mm)
Standard Drip Replenishment (mm/day)
Midday Conflict Micro-Pulse (15m ON/45m OFF)
(c) Canopy Leaf-Wetness Duration and Late-Blight Severity Progression in Treatment T4
Leaf Wetness (h/day)
Blight Severity (%)
Nocturnal Leaf Wetness Duration (h/day)
Blight-Conducive Weather Windows
Foliar Blight Severity (%)
Day of Growing Season (Days After Planting)